Nikita Kalgushkin - Backend engineer

Github | rootacces00@gmail.com | LinkedIn | Valencia/Spain | Open for remote(pref)/hybrid

Summary

Software Engineer with 7+ years of experience building scalable, data-intensive applications across telecom, retail, and EdTech sectors. Proven expertise in distributed systems, microservices architecture, and cloud infrastructure. Specialized in Python development, real-time data processing pipelines, and ML/AI integrations. Track record of optimizing system performance (1K to 5K RPS), reducing infrastructure costs, and leading cross-functional teams.

Technologies

Main language: Python; secondary languages: Kotlin, Java, Go, JavaScript

Core Expertise: search engines, NLP, Time Series, distributed computing, web, Big Data, microservices

Skills: Django, flask, FastAPI, AWS, GCP, Kafka, Redis, ElasticSearch, Clickhouse, PostgreSQL, Grafana, CI/CD, GitHub, Kubernetes, Docker, SOLID/DRY/OOP/OOD/DDD

Employment history

Software Engineer | *SymphonyAI (AI Retail)* **October 2023 - December 2025**

Software engineering services to support Store Intelligence platform with Python, AWS, Machine Learning

- Designed and implemented the Feature Vector (FV) Manager, a distributed feature vector storage leveraging ArcFace extraction, processing 1M+ FVs, for real-time product recognition and compliance. Developed advanced data pipelines, employing patterns like SAGA and product outbox to optimize vector generation and data flow across regions using AWS tools (SQS, S3, lambda), FastAPI, and PGVector extension.
- Implemented efficient data synchronization with sub-minute latency between microservices, enabling fast and reliable feature vector exchange. Simplified PostgreSQL partition management by implementing an Alembic operation to handle partition modifications and data migration(1TB+ table).

Software Engineer | *Megafon (Russian top 1 mobile operator)* **December 2022 - November 2023**

A2P Antifraud and BI system (saved the company $30K+)

- Led a team of two members (1 backend engineer and 1 data scientist), mentoring them while building the system from scratch. Set up CI/CD pipelines using **GitLab** for streamlined code quality checks and faster deployment to **Kubernetes**.
- Designed and implemented a scalable, reliable distributed microservices architecture capable of processing 200 million daily requests with a 99.9% SLI. Enhanced performance using **Kafka** for service integration, **Sentry** and **Grafana** for monitoring, **Prometheus** for metrics collection, and **ElasticSearch** for log analytics. Increased data processing speed(ETL pipelines) (1k to 5k rps) by migrating Java Kafka service to Python. Integrated **NER** for data anonymization and optimized SQL queries and APIs using **Redis** caching and indexing.

Search and ranking text engine

- Engine has 3 parts: data preprocessing(Text to Vectors), neural network and serving. Used Python, Flask, Faiss, Pytorch. Designed and built end-to-end (source code).

Software Engineer | *Funbox (online services for mobile operators)* **June 2020 - December 2022**

- Built distributed image search engine processing 200GB+ dataset using OpenCV and Spark. Developed RESTful APIs (Django) for mobile application with user profile features. Collaborated with Data Science team. Built and supported image processing pipeline for the search engine.

Fullstack Software Engineer | *Freelance* **November 2018 - June 2020**

- Implemented Warehouse, Taxi, and access management platforms with Python and Vue as a Fullstack Software Engineer in Retail and EdTech.

Education

Computer Science degree (2015-2019). Completed AI specialization on Coursera and Karpov Courses advanced ML for business covering search engines, A/B testing, dynamic pricing, and uplift modeling.